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Weak coupling of observed surface $PM_{2.5}$ in Delhi-NCR with rice crop residue burning in Punjab and Haryana

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Air pollution impacts on human health are of serious concern in northern India, and over the Delhi National Capital Region (NCR) in particular. The Kharif crop residue burning (CRB) is often blamed for degradation of Delhi-NCR's seasonal air quality. However, the concentration of fine particulate matter ($PM_{2.5}$) remained stable in Delhi, while the fire detection counts (FDCs) from satellites over Punjab and Haryana declined by 50% or more during 2015–2023. We measured $PM_{2.5}$, carbon monoxide (CO) and related parameters over Delhi-NCR, Haryana and Punjab from a network of 30 low-cost sensors (CUPI-Gs) in a selected period (September–November) of 2022 and 2023. Measured $PM_{2.5}$ showed lower concentration in 2023 compared to 2022 at Punjab and Haryana sites, in compliance with FDC reductions. Using the CUPI-G measurements, airmass trajectories, particle dispersion and chemical-transport model simulations, we show that the CRB emissions over Punjab contributed only a meagre ~14% to the overall $PM_{2.5}$ over Delhi-NCR during October-November 2022. This indicates that there exists only a very weak coupling between $PM_{2.5}$ mass over Delhi-NCR and the CRB over Punjab, highlighting the effectiveness of the Graded Response Action Plan (GRAP) in controlling air pollution in the region.

Air pollution and human health have been critical issues for developing nations and rapidly growing cities worldwide, with Delhi frequently topping the list of the most polluted cities in the world in recent years[1]. Chronic obstructive pulmonary disease (COPD), stroke, ischaemic heart disease (IHD), lung cancer, and pneumonia are the specific health outcomes most strongly linked with exposure to particulate matter with an aerodynamic diameter of less than 2.5 micrometres ($PM_{2.5}$)[2–8]. In 2019, India accounted for 1.67 million deaths attributable to air pollution and reduction in disability-adjusted life-years (DALYs) has costed 1.36% of India's gross domestic product (GDP)[9].

Delhi's geographic and climatic affliction, combined with elevated levels of anthropogenic emissions, keeps air pollution ($PM_{2.5}$) at a severe level, particularly in the autumn season[10–13]. In this context, the National Clean Air Programme (NCAP) was launched to develop a multi-sectoral policy to reduce emissions, enhance monitoring mechanisms and promote public awareness[14–17]. To improve air quality, India introduced Bharat Stage (BS)-I as the country's first attempt at transitioning to cleaner fuel in 2000, followed by BS-II, BS-III, BS-IV and BS-VI in 2001, 2005, 2010 and 2020, respectively[18]. Later, the Commission for Air Quality Management (CAQM) introduced Graded Response Action Plan (GRAP) Stages on the Delhi-NCR to control the air quality on an emergency basis[19]. Apparently, the socio-economic developments in Delhi-NCR have outpaced the air pollution control measures in the past quarter of a century.

The harvesting season of kharif rice (October-November) coincides with changing weather patterns from southeasterly trade winds to north-westerly, favouring air pollutants stagnation due to the start of the winter season. It has also been noticed that crop residue burning (CRB) is further intensified in a narrow time window for field preparation for rabi crops, as kharif rice harvesting is delayed by the Preservation of Subsoil Water Act in 2009[20–22]. Numerical modelling studies, without observational data in the CRB source region, have attributed between 20 and 40% of the $PM_{2.5}$ concentration in Delhi-NCR to non-local fire emissions during October-November[23–26]. Most of the CPCB sites are located over the city/urban areas, and the satellite retrievals fail under the smoky or cloudy conditions[10]. The

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association of CRB in Punjab and Haryana with rising $PM_{2.5}$ in Delhi-NCR is not clearly established because of the uncertainties in pollutant emissions from the source and receptor regions[10]. Thus, Delhi-NCR's air pollution mitigation policy probably over relied on the reduction of CRB emissions[27]. The CRB inventory emissions are estimated to be 15 and 172 Gg for $PM_{2.5}$ and CO, respectively, as per the Global Fire Assimilation System (GFAS) for the Indo-Gangetic Plain (Punjab, Haryana, Uttar Pradesh, and Bihar) during October-December 2022[28] or boosted up to 70 and 800 Gg, respectively, by accounting for missing fires[27]. The anthropogenic emissions for the same area and months[27] in 2018 are 300 and 4000 Gg for $PM_{2.5}$ and CO, respectively, by the hemispheric transport of air pollution (HTAP)[29]. Further, region-specific analysis suggested up to 137 and 57 Gg of $PM_{2.5}$ emissions from Punjab and Haryana alone from whole-year CRB[30].

Under these circumstances, we aimed to carry out measurements from Punjab, Haryana, and Delhi-NCR from uniformly spaced locations (Fig. S1) and performed model simulations to better understand the linkages between CRB and state-level $PM_{2.5}$, followed by inter-state coupling of air pollution.

Results and discussion

$PM_{2.5}$ and fire detection counts

The active fire detection counts (FDCs), from the Moderate Resolution Imaging Spectroradiometer (MODIS; since 2002) and the Visible Infrared Imaging Radiometer Suite (VIIRS; since 2012), are considered as a proxy for open CRB during the post-harvesting season over Punjab and Haryana. FDCs show a persistent decline in the number of CRB since 2016 for both Punjab and Haryana since the high-quality measurements of $PM_{2.5}$ are started at the US Embassy, New Delhi (Fig. 1 and Fig. S2). Between the first (2015–2016) and last (2022–2023) 2 years of the $PM_{2.5}$ measurements, the seasonal (September–November) total FDCs for Punjab and Haryana have declined by ~49 and ~72%, respectively, and drastically by ~31 and ~37% in 2023 compared to 2022. Such reductions are achieved by the law-and-order departments of the state and central governments, as well as access to facilities/crop residue machines for farmers to process crop residues without burning[31,32]. The monthly mean MODIS FDCs exhibit a delay in peak FDC by up to 2 weeks during four-time intervals (2002–2008; 2009–2012; 2013–2017; 2018–2022) (Fig. S2c, d), a result of delayed planting and harvesting following the Preservation of Subsoil Water Act[20]. The National Green Tribunal Act (2010) and its amendment in 2016 passed an order to ban CRB in northwestern India and allocated funds for compensation to the farmers in return[33].

However, as discussed in our earlier study using the measurements for 2022[10], the association of CRB in Punjab and Haryana with $PM_{2.5}$ air pollution in the Delhi-NCR was not very clear, except for two major events during 2–3 and 10–11 November 2022, while prolonged periods of high $PM_{2.5}$ (exceeding 200 $\mu g\ m^{-3}$) were observed in the southwest Punjab and western Haryana. We analysed trends in $PM_{2.5}$ measurement from the US embassy, New Delhi, which has the longest record for high-quality measurements (2015-continuing), for understanding co-evolution of CRB in the rural areas and $PM_{2.5}$ in Delhi-NCR (Fig. 1). September–November mean $PM_{2.5}$ at US embassy for 9 years (2015–2023) showed a statistically significant correlation with the FDC of Punjab and Haryana (Pearson's moment correlation coefficient, $r = 0.35$ ($p = 0.015$) and $r = 0.65$ ($p = 0.001$), respectively). The correlation may arise from similarity in the region's climate conditions, e.g., drying soil and reversal of trade winds in the post-monsoon season or due to the association of $PM_{2.5}$ to CRB in the region.

Nevertheless, the $PM_{2.5}$ decrease of 24% at the US embassy over the initial and final two years of measurements is considerably less than the reductions seen in FDCs in Punjab (49%) and Haryana (72%). Further on the interannual scale, we find an increase (20%) in $PM_{2.5}$ at the US Embassy from 2022 to 2023, while the FDCs decreased significantly (31–37%). These observations raise doubt in our understanding of how CRB directly affects the air pollution of Delhi-NCR when weighed against the controls of local emission sources from industrial/social activities and infrastructure developments in Delhi-NCR. Since our analysis is for Sep-Nov months, the overall trend is not affected by the COVID-19 lockdowns in India, which were enforced during March-June 2020[34]. In the following sections, we analyse spatial distributions of $PM_{2.5}$ from a network of 30 low-cost sensors and supporting emissions and meteorology drivers for elucidating the transport efficiency of $PM_{2.5}$ from Punjab/Haryana to Delhi-NCR and local-regional air pollution mechanisms.

Spatio-temporal variabilities in hourly mean $PM_{2.5}$

To elucidate the effect of CRB or FDC in Punjab and Haryana on Delhi-NCR, we use $PM_{2.5}$ and CO observations at a network of about 30 sites during September–November of 2022 and 2023, from the campaigns operated by the Aakash Project[35]. The $PM_{2.5}$ observations of September–November 2022 showed high $PM_{2.5}$ episodes at Delhi-NCR sites on two occasions, 02-03 and 10–11 November 2022, that were clearly linked to the CRB in Punjab or west Haryana when the strong winds blew straight from Punjab to Delhi[10] (Fig. S3). The $PM_{2.5}$ (and CO) concentrations fluctuated dramatically from day to day and from site to site in both 2022 and 2023 (Fig. 2). We find delays of about 10 days in FDCs (mostly occurring after 5 November) and lower cumulative FDCs in 2023 for almost all districts of Punjab, compared to 2022 (Fig. S4). This delay in FDC occurrence could explain the higher baseline values of $PM_{2.5}$ in Punjab and, to some extent, in Haryana in November 2023 when compared to November 2022 (Fig. 2 and Table S2).

Figure 2 shows the 6-h average $PM_{2.5}$ concentrations from all Punjab sites varied from 40–100 $\mu g\ m^{-3}$ in October 2023 to 79–338 $\mu g\ m^{-3}$ in November 2023. Similarly, in Haryana, the average $PM_{2.5}$ levels ranged from 63–112 $\mu g\ m^{-3}$ in October 2023 to 136–318 $\mu g\ m^{-3}$ in November 2023. These circumstances in 2022 were different as the CRB activities started in early October. The highest $PM_{2.5}$ daily mean peaks at selected sites of Punjab (NS-04, NS-08, NS-11 and NS-10), reached up to 1146, 832, 665 and 349 $\mu g\ m^{-3}$, respectively, between 01 and 10 November 2023 (red lines in Fig. 2). These sites of Punjab are located towards the southwest Punjab which observed the highest $PM_{2.5}$ concentrations as well as the FDCs (Fig. S3). Similarly, the sites in Haryana (NS-15 and NS-19) observed $PM_{2.5}$ levels in the range of 375–450 $\mu g\ m^{-3}$.

The role of meteorology in the accumulation and transport of air pollutants

The dispersion, transformation, and concentration of air pollutants in the atmosphere is influenced by the interplay of meteorological conditions with air pollutants. We analysed maps and time series of fire counts, rainfall, and wind vectors for the CRB period of 2022 and 2023 (openly available on the Aakash Project website[35]). Both years had significant FDC over the southwest region of Punjab than other parts of the study region where daily mean $PM_{2.5}$ levels exceeded 300 $\mu g\ m^{-3}$, although the meteorological conditions have been radically different for the 01-12 November of 2022 and 2023 (Fig. S3), suggesting the region is clearly under the influence of intense CRB emissions. In November 2022, northwesterly winds prevailed, allowing the transport of airmass from Punjab and Haryana to Delhi-NCR on two occasions[10]. The southwesterly low-wind condition (<1 $m\ s^{-1}$) restricted the movement of the air mass resulting in an accumulation of local pollutants in Delhi-NCR in November 2023.

To track the movement of air masses more precisely over a 24-h period, we used backward trajectories for each day of the CRB period (September–November of 2022 and 2023) at three different heights (200, 500 and 800 m) from the hybrid single-particle Lagrangian integrated trajectory (HYSPLIT) model[36]. The structures of the trajectories for 2022 and 2023 at 500 m are distinct across all CUPI-G sites segmented into four regions: North Punjab, Southwest Punjab, Central Haryana, and Delhi-NCR (Fig. S5). In September (2022 and 2023), pollutants at Punjab and Haryana stations arrived primarily from the northwest region, but Delhi-NCR sites received pollutants from the southeast (the Uttar Pradesh region). The month of November in 2022 saw a substantial influence of pollution from the northwest over Delhi-NCR, which was relatively low in November 2023 (Fig. S5, bottom-right panel). This, reaffirmed that the air remained

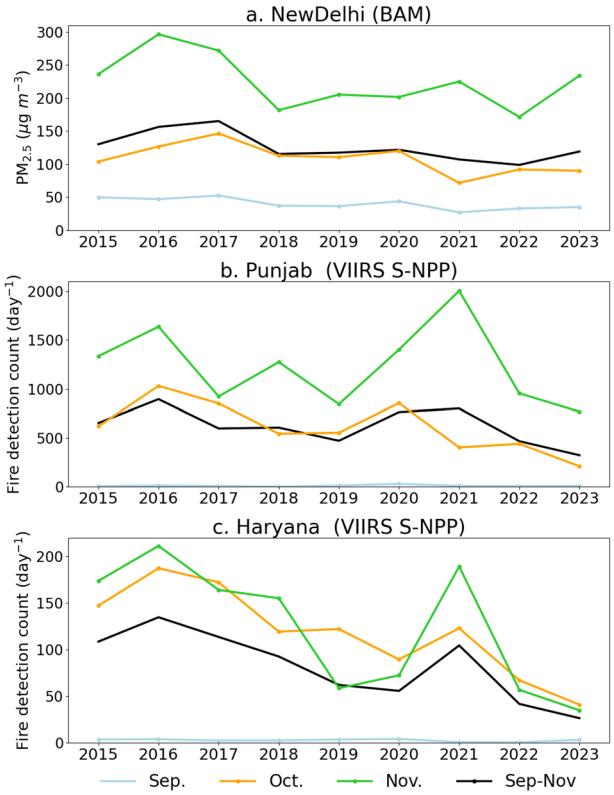


Fig. 1 | Changes in $PM_{2.5}$ in Delhi and FDC in Punjab and Haryana. a Time series of monthly and seasonal mean $PM_{2.5}$ measured at the United States of America (US) embassy, New Delhi and (**b**) VIIRS fire detection counts (FDCs) in the neighbouring states of Punjab and (**c**) Haryana in the period of 2015–2023. The $PM_{2.5}$ measurements at the US Embassy are made using the beta attenuation mass (BAM) monitor, which obtained the US EPA Federal Equivalent Method (FEM) designation for continuous $PM_{2.5}$ monitoring. The daily total VIIRS FDCs are shown as mean over the months of September to November, covering the full CRB season. Both the MODIS and VIIRS show closely matched trends and variability in FDCs, but the absolute counts are about five times greater for VIIRS than MODIS, mainly due to smaller fire pixel size in the case of VIIRS (Fig. S2).

stagnant (low-wind condition), contributing to the extremely high $PM_{2.5}$ dome over Delhi-NCR in the October-November months of 2023.

$PM_{2.5}$ levels amidst policy interventions in Delhi-NCR

Delhi-NCR's AQI (Air Quality Index) start to deteriorate in October, and a systematic rise in $PM_{2.5}$ through November occurs every year in north-western India due to changing meteorology, including the minimal rainfall and slow winds (Fig. S3), lowering of boundary layer (Fig. S6). The CAQM implemented the GRAP Stages to regulate air quality by restricting the activities of local sources—an option that is guaranteed to work from recent past experiences[37,38]. The GRAP is categorised into Stages I, II, III and IV, when the AQI ranges from 201 to 300 (poor), 301 to 400 (very poor), 401 to 450 (severe) and greater than 450 (severe plus), respectively[19]. The commencement dates of the GRAP Stages in 2022 and 2023 were not so different (Fig. 3). The implementation of GRAP Stage III/IV coincided with high pollution events in Delhi-NCR, when $PM_{2.5}$ in Delhi-NCR exceeded 400 μg m^{-3}, as observed by CUPI-G network in both years. GRAP IV was rather in effect for a shorter duration in 2022, as there was a noticeable improvement in $PM_{2.5}$ air quality because wind speed increased to over 2 m s^{-1} in November (Fig. 3c). Because the baseline values of $PM_{2.5}$ remained 30% higher in November 2023 than November 2022, GRAP IV continued for an extended period in 2023, under very low-wind speed of ~1 m s^{-1} (Fig. 3d; bottom row).

With the enforcement of GRAP III, followed by GRAP IV, our measurements showed a steady decline in $PM_{2.5}$ afterwards (Fig. 3a, b). The $PM_{2.5}$ at NS-23, NS-28 and NS-30, reaching highest levels in 2023 (553, 339 and 438 μg m^{-3}, respectively) before GRAP IV (2–5 November), dropped to 360, 234 and 276 μg m^{-3}, respectively (6-8 November). A similar drop in $PM_{2.5}$ levels can also be observed in 2022 during the GRAP IV period (Fig.3b). Thus, the decrease of PM2.5 levels in Delhi-NCR was mainly attributed to GRAP IV when the major $PM_{2.5}$ emissions from road traffic and construction activities, among other sources, were reduced. However, a rise in $PM_{2.5}$ load was recorded after the revocation of GRAP IV for 2022 and 2023. This analysis suggests a close link of mean $PM_{2.5}$ levels with wind speed and BLH (both inversely related) and low dependency on wind direction (Fig. 3; bottom row).

Diurnal and synoptic variability of $PM_{2.5}$ and CO over the study domain

We studied the diurnal variation in two markers of air pollution ($PM_{2.5}$ and CO) for the period of medium to intense burning (01 October–30 November) across all CUPI-G sites (Fig. 4 and Fig. S7). Plots of individual sites are available on Aakash project website (https://aakash-rihn.org/en/data-set). The Delhi-NCR had night-time CO build-up from 26 October–09 November 2023, while $PM_{2.5}$ levels were high in Southwest Punjab, Central Haryana, and to a lesser degree in North Punjab (Fig. 4). The rainfall event of 10 November 2023 in Delhi-NCR brought the $PM_{2.5}$ concentration down immediately to as low as 50–60 μg m^{-3}; however, the concentration of $PM_{2.5}$ increased dramatically in the evening (night) of Diwali (12 November 2023). This vividly illustrated the impact of Diwali firecrackers on air pollution[11,39]. Because CO is a longer-lived gaseous species than $PM_{2.5}$, we would anticipate more consistent CO levels in the region if identical sources of biomass burning emissions were causing regional air pollution. There must be large common sources of CO and $PM_{2.5}$ within the periphery of Delhi-NCR those are turned off by rainwater; the behaviour of sudden and synchronised concentration changes cannot be accounted for a major fraction by direct CRB emissions farther away[11,40].

The day-night differences in the $PM_{2.5}$ is not large in the Delhi-NCR (differences within ~20%, = [night-day]/day × 100; Fig. 4d and Fig. S7d) because the $PM_{2.5}$ are probably emitted from infrastructure developmental (construction/demolition) activities during the day within the city area, when the BLH is high (Fig. S6). But the average CO concentrations are about 67% greater in the night compared to the day in 2023 (Fig. 4d), and slightly lower day-night differences (~48%) are observed for CO in 2022 (Fig. S7d). This lower difference in 2022 is caused by lower differences in BLH and higher wind speed, compared to 2023 (Fig. S6). CO in Delhi-NCR is predominantly emitted from fossil fuel combustion or local biomass burning, independent of the diurnal cycle in $PM_{2.5}$ emission sources. These fossil fuels are extensively used in road transport (petrol, diesel, CNG) and local biomass burning from agricultural residue, firewood, and organic materials for cooking or heating[41,42]. These activities emit $PM_{2.5}$, CO, NO_x, NH_3 etc.[43]. This situation is dramatically different in rural Punjab and Haryana—a clear day-night variation is observed only during the intense CRB periods, i.e., from 29 October to 25 November in Southwest Punjab (Fig. 4b), 20 October to 16 November in Central Haryana (Fig. 4c) and 23 October to 10 November in North Punjab (Fig. 4a). The day-night differences in $PM_{2.5}$ are greater than those in CO in the CRB dominated areas suggesting that open CRBs are lower in CO emission and higher in $PM_{2.5}$ emission compared to the industrial sources (Table S3).

Sensitivity of $PM_{2.5}$ variations in Delhi-NCR to emissions from CRB

The major air pollutants are simulated using the Community Multiscale Air Quality Modelling System (CMAQ) with control emissions (see Methods)[44]. Because of the coarse horizontal resolution (30 × 30 km) of the model compared to the distance between some of the sites, we show the regionally averaged observation and model comparisons of $PM_{2.5}$ and CO for 2022 (Fig. 5). The control simulations clearly underestimated $PM_{2.5}$ at all regions in the period of 15 October through the end of November, particularly for the period of CRB in Punjab and Haryana (Fig. 5). This suggests that the "primary" emissions of $PM_{2.5}$ by the global fire assimilation system (GFAS) emissions are

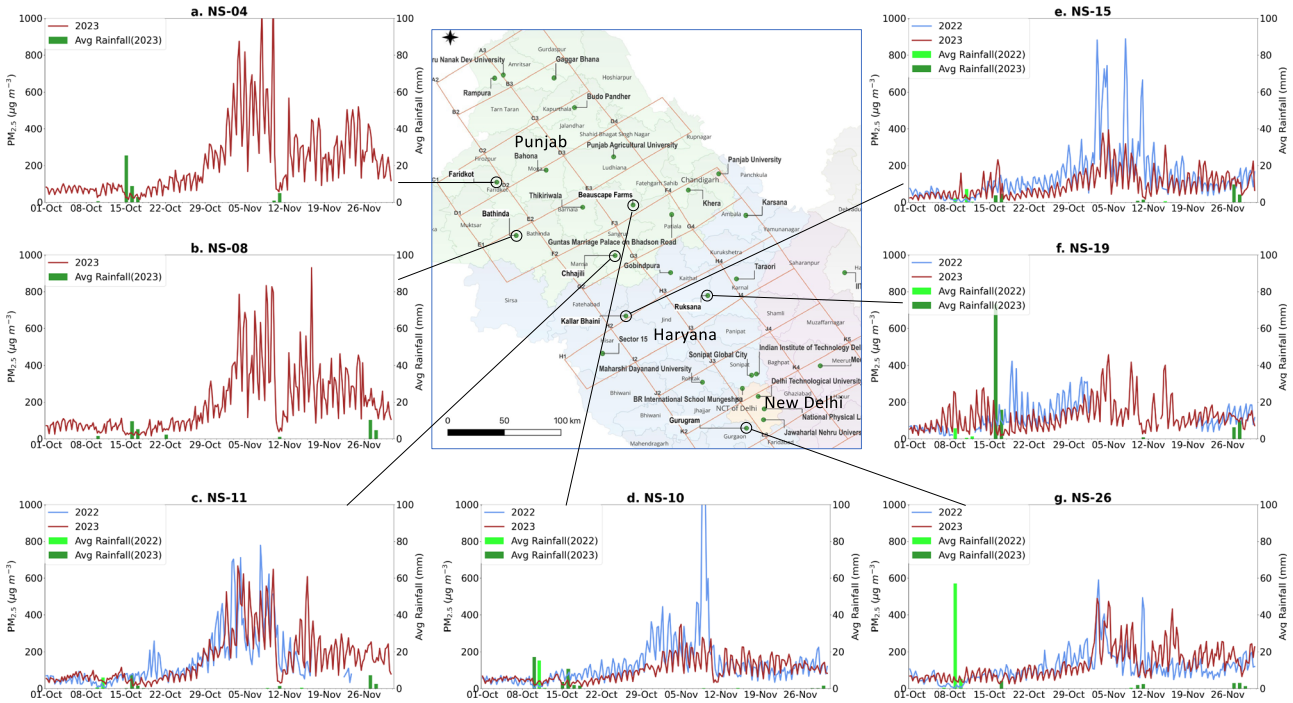


Fig. 2 | Observed variations in $PM_{2.5}$ and rainfall in our study region. We show 6-hourly mean $PM_{2.5}$ at seven selected sites at Faridkot (**a**, NS-04), Bathinda (**b**, NS-08), Chhajili (**c**, NS-11) and Beauscape Farms (**d**, NS-10) in Punjab, at Kallar Bhaini (**e**, NS-15) and Ruksana (**f**, NS-19) in Haryana, and at Gurugram (**g**, NS-26) in Delhi-NCR during 01 October to 30 November of 2022 (as per site availability; blue lines) and 2023 (red lines). The daily rainfall events are marked by light and dark-green-coloured bars for 2022 and 2023, respectively. The site codes, arranged from northeast to southwest of the CUPI-G network, and names mapping are given in Fig. S1 and Table S1.

not sufficient to explain the observed concentrations (Tables S4, S5). The uncertainty in emissions of various air pollutant species is still large because of different methodologies for calculation of burning fraction and fuel load to estimate the fire emissions[45]. Nonetheless, the SOA formation from CRBs and anthropogenic sources could also be inadequate in CMAQ simulations (details will be addressed in future studies). The emissions in GFAS and chemistry in CMAQ are well represented for explaining the observed spatial and temporal variabilities in CO during the CRB season of 2022.

However, a CMAQ sensitivity simulation by varying primary $PM_{2.5}$ emissions from CRB (Emis_8x; see Methods) could help us to quantify $PM_{2.5}$ variability in Delhi-NCR that is contributed by the "direct" emissions of CRB in Punjab and Haryana. The Emis_8x simulation shows increase in $PM_{2.5}$ concentrations in Punjab, to the level of observed $PM_{2.5}$ from 14 to 30 October over north Punjab and from 20 October to 18 November over southwest Punjab (Fig. 5). However, we do not find the eight times linear increase for $PM_{2.5}$ in Delhi-NCR, because of the shorter lifetime of $PM_{2.5}$ and heterogeneity of the primary emissions being smeared out in coarse CMAQ resolution. For the eight times high CRB emissions, we find a marginal increase in $PM_{2.5}$ at the surface sites in central Haryana, except for two peak events on 03 November and 12 November 2022 (Fig. 5c). The partial obscuration of the satellite field of view by cloud cover during 02–04 November 2022 led to reduced emission estimation from the Punjab and Haryana in the control case[10]. However, during 08–10 November 2022, the cloud cover was extensive over the region and little $PM_{2.5}$ increases were achieved for southwest Punjab or central Haryana even for the Emis_8x case. The limitation of satellite remote sensing in cloud cover conditions highlights the importance of continuous measurements from a dense observation network on the ground that can only be achieved by using low-cost and low-maintenance sensors.

We estimated the direct emissions of $PM_{2.5}$ (01 October–30 November) from Punjab to Delhi-NCR (14.4%; = $[(\text{control} - \text{Emis_8x})_{\text{Delhi-NCR}}/(\text{control} - \text{Emis_8x})_{\text{Punjab}} \times 100]$ and similarly from Haryana to Delhi-NCR (44%), by using the two CMAQ simulation cases (Fig. 5d). One could argue that the Haryana to Delhi-NCR contribution contains large double counting because the CRB emissions of Punjab are affecting the $PM_{2.5}$ increase in Haryana first, before reaching Delhi-NCR, particularly evident during the two peak events of 2022[10] (Fig. 5c). This is a possible explanation for the high correlation ($r = 0.65$) of the $PM_{2.5}$ at the US embassy with Haryana FDC due to its proximity to Delhi-NCR. The CRB emission signals from Punjab move in different directions (Fig. S3), but those reached Delhi-NCR must travel over Haryana. Note here that FDCs in Haryana are less by an order of magnitude than those observed in Punjab during 2022 and 2023 (Fig. 1), and occur only in the western boundary of the state of Haryana (Fig. S3). Thus, a negligible/small contribution of CRB in Haryana to Delhi-NCR's $PM_{2.5}$ in 2022 is expected. Further analyses using measurements and models are needed for more accurate attribution.

Our study exemplifies that the amount of biomass burned in Punjab and Haryana does not always determine the air quality of heavily urbanised Delhi-NCR, even during the core rice-stubble burning period (October-November). Our measurements show the persistence of elevated levels (150 $\mu g\ m^{-3}$ or greater) of $PM_{2.5}$ in the Delhi-NCR until February of each year (not discussed here). Human health and climate risks of high $PM_{2.5}$ (aerosols) and pollutant gases must be better studied at the major urban centres and rural areas, using uniformly distributed observational networks. Because the transport of air masses transcends geopolitical boundaries across various regions, wider cooperation within and among the countries on the subcontinent is strongly recommended to overcome seasonal and annual persistent air pollution. We also propose that the solutions must be framed for eliminating CRBs that would benefit the reduction of local air pollution exposure and facilitate the climate goals by reducing greenhouse gas emissions.

Methods

Compact and useful $PM_{2.5}$ instruments with gas sensors (CUPI-G)

The CUPI-Gs consist of various sensors to continuously monitor fine particulate matter ($PM_{2.5}$), temperature (T), relative humidity (RH), GPS signal (coordinates and time), and several key air pollutants such as carbon monoxide (CO), nitric oxide (NO), nitrogen dioxides (NO_2), potential

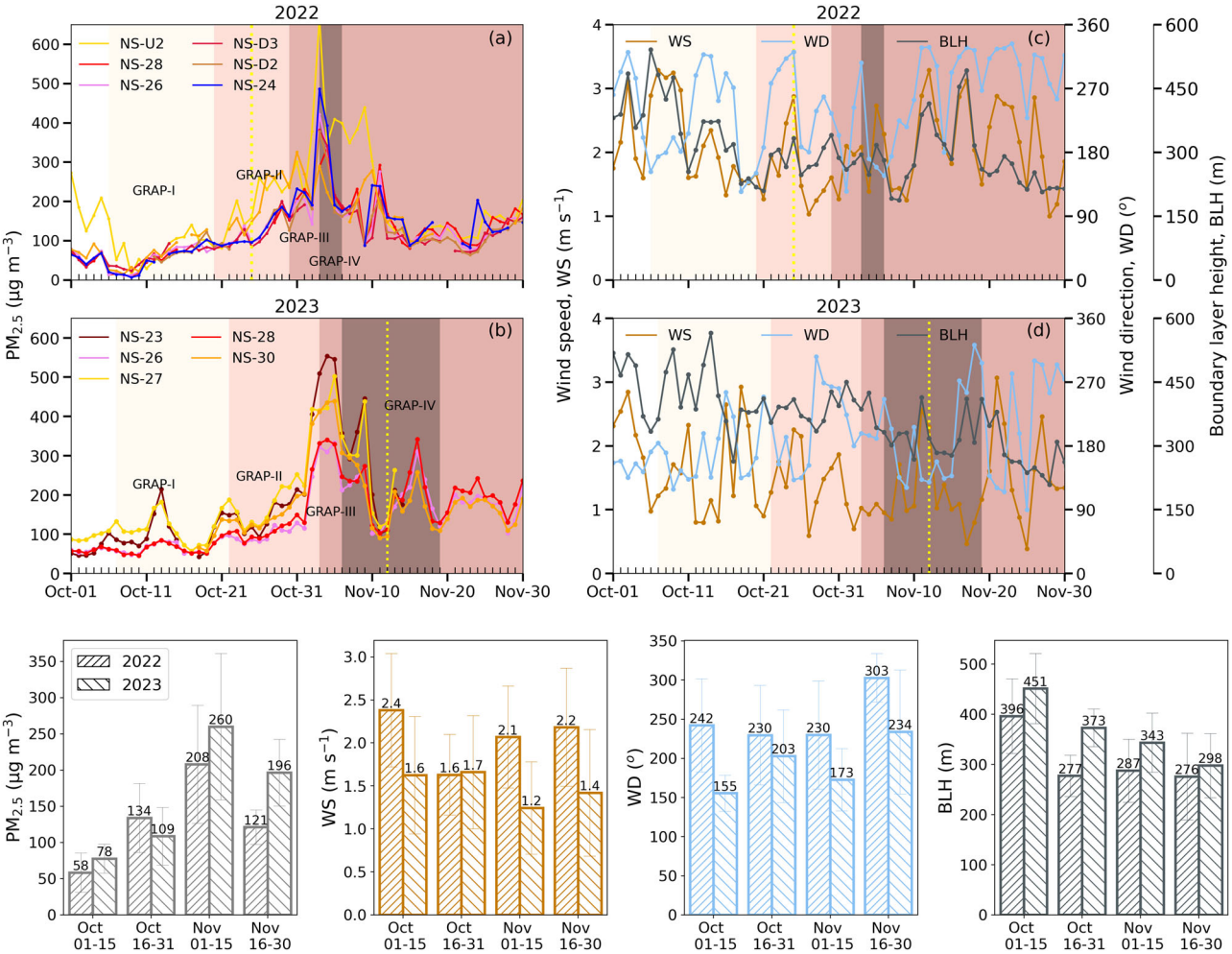


Fig. 3 | Controls of meteorology and emission mitigation on $PM_{2.5}$ in Delhi-NCR. Daily average $PM_{2.5}$ concentration over Delhi-NCR in 2022 (NS-28: JNU, NS-U2: Civil Lines, NS-D3: Sohna, NS-D2: Faridabad, NS-U3: Aurangabad, NS-24: Sonipat Global City, NS-26: Gurugram) (**a**), and in 2023 (NS-28: JNU, NS-26: Gurugram, NS-23: Sonipat (IIT-D), NS-27: DTU, NS-30: Meerut) (**b**). Wind speed, wind direction (in degree; 0° denotes northerly winds) and daily average planetary boundary layer height shown for 2022 (**c**), for 2023 (**d**). The yellow dotted line is a marker for Diwali day, and the shaded colour blocks represent the starts and terminations of the GRAP Stages for both years. The bottom row shows the mean values of $PM_{2.5}$, WS, WD and BLH (panels arranged: left to right) for 2022 and 2023 (marked by patterns within the bars).

ozone (combined O_3 and NO_2)[46]. Details of each sensor are given in (Table 1). Additional information on sensitivity, quality control (QC), and quality assessment (QA) of the gas sensors may be accessed at https://www.alphasense.com[47]. The $PM_{2.5}$ sensor uses light scattering measurements to calculate mass concentration. The $PM_{2.5}$ sensors are calibrated and validated at the laboratories of Nagoya University and Nagasaki University, prior to infield deployment. The instrumental output is compared with those from the beta-ray attenuation method (BAM), and calibration scaling factors are established for each of the $PM_{2.5}$ sensors. During the campaign, the equipment was not periodically calibrated; instead, we monitored its condition by comparing the readings of CUPI-Gs with that of instruments in proximity, e.g., a BAM instrument at the US Embassy, New Delhi, and the baseline behaviour through online monitoring.

The measurements were conducted continuously around the year at several sites, and for more than six months at most of the 30 locations in a year during 2020–2023. However, we present data for the 01 Sept–30 November only here for the Kharif CRB season. The measurement data were collected and saved in near real-time every 2 min, and uploaded to an internet server once a day. If the measurement was suspended for more than 4 min (due to power shortage), data from the first 5 min after resumption were not used. If erroneous data were discovered, the data were cleansed and subjected to further processing (e.g., ref. 10). The data collection efficiency was 97% for $PM_{2.5}$ and 70% for CO. Data was recorded every 2 min, and any invalid or erroneous data were removed. Daily averages were calculated only if over 50% of the 2-min data were valid. We have recorded 90 and 65% of valid hourly data for $PM_{2.5}$ and CO, respectively, after removal of the erroneous data. After QA/QC, we prepared hourly mean data for final analysis. The CUPI-G $PM_{2.5}$ variabilities at 4–5 sites in Delhi-NCR show significant correlation with those at the US Embassy (mean $r = 0.94 \pm 0.04$; mean slope = 1.01 ± 0.21), even though the locations have their own local characteristics (Fig. S9). Whereas similar tests were difficult for CO measurements by CUPI-G because of the lack of high-quality CO data in Delhi-NCR (at the US Embassy or elsewhere). However, we have checked the measurements at repeating CUPI-G sites in 2023 which showed excellent repeatability with respect to the 2022 measurements before and after the CRB season (late August to early September and after late November) for both $PM_{2.5}$ and CO. These observations have raised our confidence on the QA/QC performed on the low-cost sensors (dated: 05 June 2024, at Nagasaki University) before field deployment, and suitability for the analysis that are presented here. Sensitivities of both $PM_{2.5}$ and CO sensors were tested in Nagasaki, Japan, by comparing with standard instruments based on BAM monitor and non-disperse infrared spectroscopy, respectively, and their unit-to-unit variations were estimated to be 29 and 21%, respectively (Nakayama et al., in prep.).

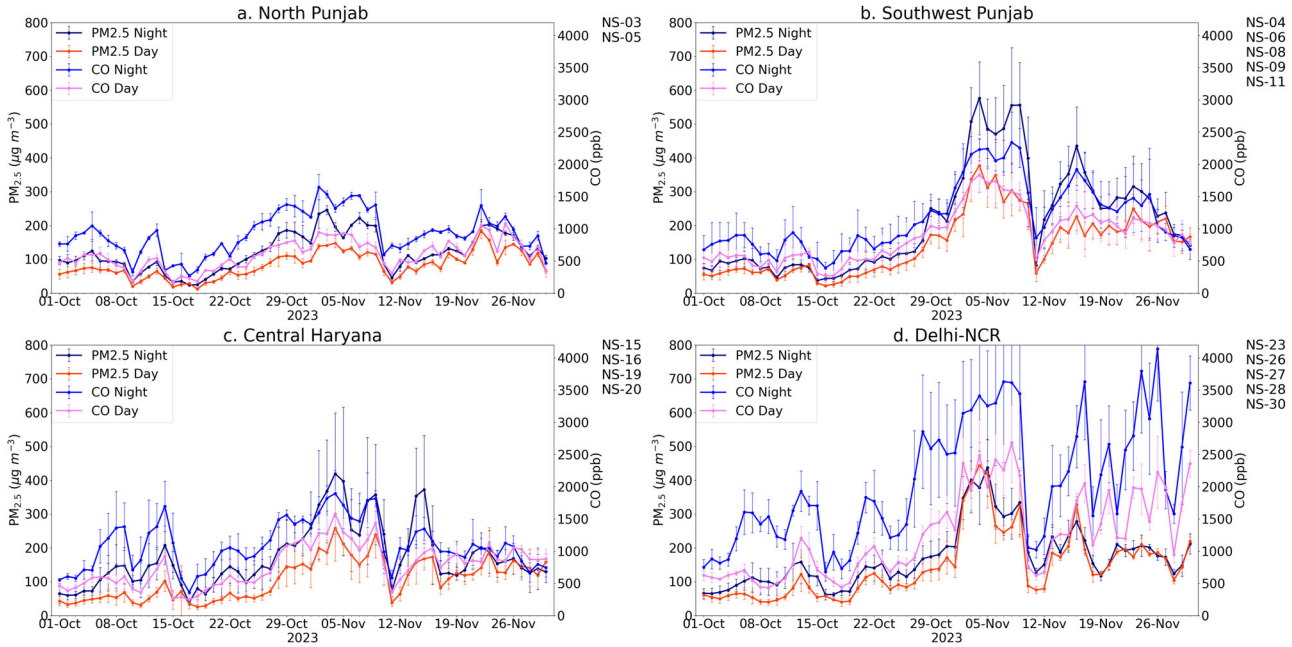


Fig. 4 | Day-night variations of $PM_{2.5}$ and CO in 4 subregions of our study. Day (0600–1800 IST) and night (1800–0600 IST) time average values of $PM_{2.5}$ and CO at selected CUPI-G sites of (**a**) North and (**b**) Southwest Punjab, (**c**) Central Haryana and (**d**) Delhi-NCR measured in 2023. Site selections are performed based on similarity in variability of $PM_{2.5}$ and CO concentrations, and minimal data gaps due to instrumental operation. Similar results using 2022 campaign data are shown in Fig. S7. The day and night-time average BLH plots are shown in Fig. S6.

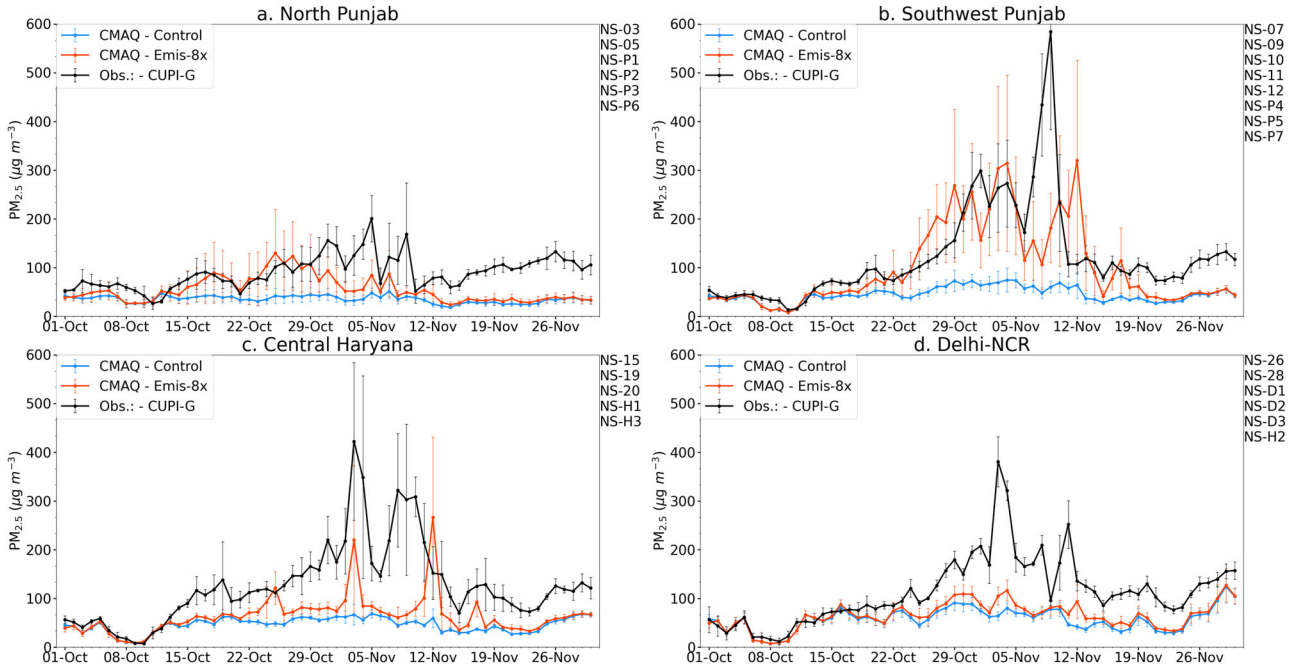


Fig. 5 | Model-observation comparisons of $PM_{2.5}$ and model sensitivity to CRB emissions. Comparisons of $PM_{2.5}$ measured (black line) at the CUPI-G sites in 2022 and WRF-CMAQ simulations with two different emission intensities (blue: control case, orange: Emis-8x, eight times more emissions from CRB only relative to the control case). The error bars indicate the site to site variation for a varied number of sites within the region (see legends). Site-specific variations for 2022 are available on the Aakash project website. Results are shown for 4 subregions of our study: North Punjab (**a**), Southwest Punjab (**b**), Central Haryana (**c**), and Delhi-NCR (**d**).

CUPI-G monitoring site networks in 2022 and 2023

We installed CUPI-G instruments at strategically located sites in northwest India, covering the rural, semi-rural, and urban regions of Punjab, Haryana, Delhi-NCR and western Uttar Pradesh. Measurements capture the concentrations of $PM_{2.5}$, CO, NO_x, O_3, and meteorological variables that are of critical importance to understand the emission, transport, and transformation of major air pollutants that originate from CRB. Our strategy is unique in that numerous sensors were situated in rural areas rather than urban ones. Target areas were chosen based on the prevalent northwest winds from Punjab to Delhi-NCR. In 2022, 30 instruments were placed in a gridded manner[10]. Based on 2022 experiences we relocated some of the instruments to the southern side of Punjab (Faridkot and Bathinda)

Table 1 | Details of $PM_{2.5}$ and gas sensors fitted to each of the CUPI-G unit

Parameter	Principle	Sensor name	Company name	Reference
$PM_{2.5}$	Optical	Panasonic $PM_{2.5}$ sensor	Panasonic-Nagoya Univ.	Nakayama et al. (2018)[46]
CO	Electrochemical	CO-B4	Alphasense	www.alphasense.com[47]
NO	Electrochemical	NO-B4	Alphasense	
NO_2	Electrochemical	NO_2-B43F	Alphasense	
Ox	Electrochemical	OX-B431	Alphasense	

and Haryana (Sector 15, Hisar) and to the northeast side of our study area (Roorkee). The analysis of the 2022 measurement suggested that a large amount of airmass trajectories passed through the southern side of our study area and arriving in Rajasthan[10]. Along with that, the districts of Faridkot and Bathinda also experienced large fire detection counts (FDC). Most of the sites have been repeated in 2023, so that a comparative analysis of air pollutants and FDC could be performed jointly for 2 consecutive years (Fig. S1 and Table S1).

Meteorological parameters

The Global Forecasting System (GFS) 0.25° meteorological datasets are used to perform the backward and forward trajectory analysis using the HYSPLIT model version 5.2. European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) hourly data 0.25° × 0.25° datasets were used for wind vectors and BLH. A satellite-based dataset from the National Aeronautics and Space Administration (NASA) Suomi National Polar Satellite's VIIRS at 375 m resolution and 13:30 local transit time, as well as the EOS-Terra/Aqua satellite's MODIS dataset at 1 km resolution and 10:30 and 13:30 local transit times, respectively were used for fire detections. The rainfall data used in this study are taken from the Indian Meteorological Department (IMD) Daily Merged Satellite Gauge Rainfall at 0.25° × 0.25° grid resolution[48]. It is derived from global precipitation measurement (GPM) satellite rainfall merged with accumulated daily rain-gauge data (valid at 0300 UTC) of IMD stations. Data were available from the India Meteorological Department[49].

WRF-CMAQ model simulations

Simulation of air chemistry is performed using the Weather Research and Forecasting (WRF) version 4.4.1, coupled with the Community Multiscale Air Quality model (CMAQ) version 5.3.3. The model is formulated for the Indian region with a 30 × 30 km horizontal resolution, and the 37 sigma-pressure layers from the surface to 50 hPa, with the first layer at ~50 m height. The gas and aerosol modules used in the CMAQ Chemical-Transport Model (CCTM) were State Air Pollution Research Center Mechanism version 2007 (SAPRC-07)[50] and CMAQ aerosol model version 7 (Aero7)[51], respectively.

The model control simulation used anthropogenic emissions from HTAPv2 database[29] and open biomass burning emissions from GFASv1.2 based on fire radiative power at 0.1° × 0.1° horizontal resolution[28] (simulation referred to as 'control'). The sensitivity simulation was performed by arbitrarily increasing the GFAS emissions by eight times (referred to as 'Emis_8x'). The control and Emis_8x simulations cover the full range of differences between emission inventories (Tables S4, S5). Model simulations are sampled at hourly mean time intervals and the nearest grid to the measurement latitude-longitude at model level 1. The hourly mean CMAQ simulations were used to obtain the daily average values for each CUPI-G location. The daily average values of the locations were then used to determine the average value for each region.

Data availability

The datasets generated and/or analysed during the current study are available in the RIHN Aakash repository [https://aakash-rihn.org/en/dataset/]. We will mirror these datasets at the living repository following the publication of the article [https://chikyu.repo.nii.ac.jp/].

Code availability

HYSPLIT, CMAQ and WRF are open source software. The data analysis in the present study was carried out using open source software (Python), using generic tools. Codes can be made available by contacting the corresponding authors.

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Author contributions

Y.M., T.N., H.A., T.S., A.B. and Po.M. performed field observations (acknowledging the support of Takayuki Yamasaki, Nagoya Univ.); S.H. and P.K.P. managed the project; Y.M., T.N. and H.A. managed the instruments and data QA/QC; K.Y. performed model simulations in consultation with M.T. and M.K.; A.S., A.P.D., S.K.D., M.S.B., Sa.M., K.R., S.B., V.J.V., R.K.K., T.K.M., Pr.M., K.V. and Su.M. helped CUPI-G site maintenance. H.A., A.B., Po.M., P.K.P., Y.M. and N.Y. analyzed and interpreted the data. Po.M. and P.K.P. integrated all analyses and led manuscript preparation. All authors discussed the results, contributed to manuscript preparation and approved the final manuscript.

Competing interests
The authors declare no financial or non-financial competing interests.

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